GlycoTwin architecture

Designed to run 24/7 for the doctor. The prototype replays a stored 14-day trace of synthetic data. Team Aeternum.

1 DATA (synthetic)

Synthea clinic record

age, sex, BMI, HbA1c, diabetes duration
blood pressure, hypertension, kidney disease
metformin, sulfonylurea, insulin on record

Simulated wearable trace

5-minute series, seed 2026
glucose, heart rate, steps
one sleep value per night

PLANNED, not built yet

live, consented wearable / CGM
stream feeding the twin 24/7

339 synthetic patients, 14 days each

Not real watches, not lab continuous glucose

2 MODEL SERVICE (FastAPI on Render, read-only)

Join on patient id

record plus wearable trace become one table per patient

6-hour window, every 30 minutes

the model sees only the past 6 hours plus the record

LightGBM spike model

chance of reaching 180 mg/dL
within the next 2 hours

Four glucose models

glucose at 30, 60, 90, 120 min
forecast curve, display only

Alert rule

on at 60% risk or higher

Reasons for the alert

contributing factors, not causes

proxy

3 DOCTOR APP (Next.js on Vercel)

Next.js server

sign-in, then proxies every call to the API

Doctor UI in the browser

- Patient list sorted by 2-hour risk, filters, search by id
- Record first, then the glucose chart
- Same glucose, different record (148.5 mg/dL)
- 14-day replay slider
- What-if sliders, change stated in words
- Model notes: one right alert, one wrong alert
- Footer: synthetic data, not clinically validated

The browser talks only to the Next.js server

60% or higher

4 ESCALATION (designed workflow, simulation: no real message is sent)

Below 60%

no alert, the twin keeps
scoring every 30 minutes, 24/7

Tier 1: Care team

nurse or coordinator sees the alert and the reasons
contacts the patient following clinic protocol
documents the outcome

Tier 2: Doctor

only if clinic criteria are met:
symptoms, patient unreachable,
repeated alerts over days

PLANNED

designed to integrate with
ABHA and the WhatsApp
Business API

EVALUATION

People are split by person, so a test patient never appears in training. On 52 held-out patients the fused model scores AUROC 0.986, against 0.751 for glucose alone and 0.619 for the 30-minute slope. At the 60% alert: 0.272 false alarms per patient-day.

Synthetic data. The high score describes a formula-generated series and is not a clinical validation. The screen never recommends a drug dose. Dashed boxes are planned, not built.